

Class 14: RNASeq mini-project

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Background

Raw lists of genes and their information can be challenging to interpret and difficult to understand. As a result, we can use different packages in R to sift through gene databases and reduce the complexity of mapping them to known biological functions, processes and pathways. This section focuses on the HOXA1 developmental transcription factor which is required for ling fibroblast and HeLa cell cycle progression.

Data Import

Read counts and metadata CSV files

```
library(DESeq2)
```

Loading required package: S4Vectors

Warning: package 'S4Vectors' was built under R version 4.5.3

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Warning: package 'Biobase' was built under R version 4.5.3

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

# Import metadata and take a peek
colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
# Import countdata
countData = read.csv(countFile, row.names=1)
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28

ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0
ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

Sannity Check

Q. Complete the code below to remove the troublesome first column from countData:

```
countData = as.matrix(countData[, -1])
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

Setup DESeq object and Run DESeq Analysis pipeline

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

```
countData = countData[rowSums(countData) > 0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

How many genes are left after filtering?

```
dim(countData)
```

```
[1] 15975      6
```

Extract the results

Big Table with log2 fold changes and p-values

```
dds = DESeqDataSetFromMatrix(countData= countData,  
                              colData= colData,  
                              design= ~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds = DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res <- results(dds)  
dds
```

```

class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
               ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor

```

res

```

log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 15975 rows and 6 columns

```

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.9136	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
...	...	...	...	...	...
ENSG00000273748	35.30265	0.674387	0.303666	2.220817	2.63633e-02
ENSG00000278817	2.42302	-0.388988	1.130394	-0.344117	7.30758e-01
ENSG00000278384	1.10180	0.332991	1.660261	0.200565	8.41039e-01
ENSG00000276345	73.64496	-0.356181	0.207716	-1.714752	8.63908e-02
ENSG00000271254	181.59590	-0.609667	0.141320	-4.314071	1.60276e-05
	padj				
	<numeric>				
ENSG00000279457	6.86555e-01				
ENSG00000187634	5.15718e-03				
ENSG00000188976	1.76549e-35				
ENSG00000187961	1.13413e-07				
ENSG00000187583	9.19031e-01				
...	...				
ENSG00000273748	4.79091e-02				
ENSG00000278817	8.09772e-01				
ENSG00000278384	8.92654e-01				
ENSG00000276345	1.39762e-01				
ENSG00000271254	4.53648e-05				

Q: Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
low counts [2]    : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

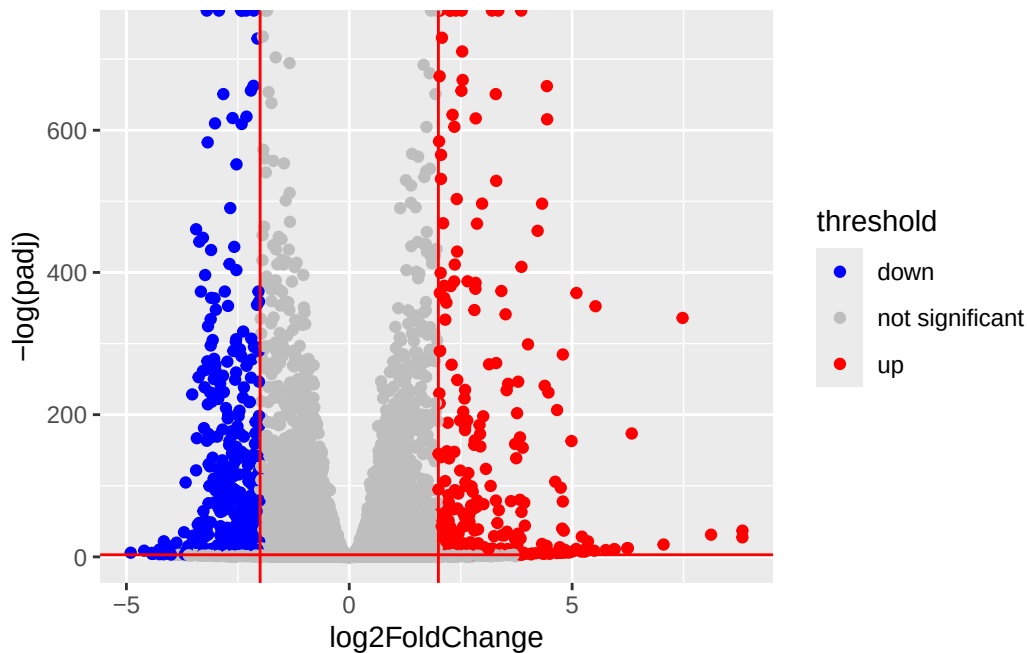
Data Viz - Volcano Plot

```
library(ggplot2)

res$threshold <- "not significant"
res$threshold[res$log2FoldChange >= 2 & res$padj <= 0.05] <- "up"
res$threshold[res$log2FoldChange <= -2 & res$padj <= 0.05] <- "down"

ggplot(res) +
  aes(log2FoldChange, -log(padj), color = threshold) +
  geom_point() +
  geom_vline(xintercept = c(-2, 2), col = "red") +
  geom_hline(yintercept = -log(0.05), col = "red") +
  scale_color_manual(values = c("down" = "blue", "not significant" = "gray", "up" = "red"))
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).



```
res <- res[order(res$padj), ]
write.csv(res, file = "deseq_results.csv")
```

Add Annotation data

Q. Use the `mapIDs()` function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

Add gene symbol and entrez ids:

```
library("AnnotationDbi")
library("org.Hs.eg.db")
```

```
columns(org.Hs.eg.db)
```

```
[1] "ACCNUM"      "ALIAS"       "ENSEMBL"     "ENSEMBLPROT" "ENSEMBLTRANS"
[6] "ENTREZID"    "ENZYME"      "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
```

```
[11] "GENETYPE"      "GO"            "GOALL"         "IPI"           "MAP"
[16] "OMIM"          "ONTOLOGY"      "ONTOLOGYALL"   "PATH"          "PFAM"
[21] "PMID"          "PROSITE"       "REFSEQ"        "SYMBOL"        "UCSCKG"
[26] "UNIPROT"
```

```
res$symbol = mapIds(org.Hs.eg.db,
  keys = row.names(res),
  keytype="ENSEMBL",
  column= "SYMBOL",
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez = mapIds(org.Hs.eg.db,
  keys=row.names(res),
  keytype="ENSEMBL",
  column="ENTREZID",
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name = mapIds(org.Hs.eg.db,
  keys=row.names(res),
  keytype="ENSEMBL",
  column="GENENAME",
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 10 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000117519	4483.63	-2.42272	0.0600016	-40.3776	0
ENSG00000183508	2053.88	3.20196	0.0724172	44.2154	0
ENSG00000159176	5692.46	-2.31374	0.0575534	-40.2016	0
ENSG00000116016	4423.95	-1.88802	0.0431680	-43.7366	0

ENSG00000164251	2348.77	3.34451	0.0690718	48.4208	0
ENSG00000124766	2576.65	2.39229	0.0617086	38.7675	0
ENSG00000124762	28106.12	1.83226	0.0388966	47.1058	0
ENSG00000106366	43719.13	-1.84405	0.0419165	-43.9933	0
ENSG00000188153	2944.13	2.26608	0.0552681	41.0016	0
ENSG00000122861	28007.14	2.26253	0.0552183	40.9742	0

	padj	threshold	symbol	entrez
	<numeric>	<character>	<character>	<character>
ENSG00000117519	0	down	CNN3	1266
ENSG00000183508	0	up	TENT5C	54855
ENSG00000159176	0	down	CSRP1	1465
ENSG00000116016	0 not significant		EPAS1	2034
ENSG00000164251	0	up	F2RL1	2150
ENSG00000124766	0	up	SOX4	6659
ENSG00000124762	0 not significant		CDKN1A	1026
ENSG00000106366	0 not significant		SERPINE1	5054
ENSG00000188153	0	up	COL4A5	1287
ENSG00000122861	0	up	PLAU	5328

	name
	<character>
ENSG00000117519	calponin 3
ENSG00000183508	terminal nucleotidyl..
ENSG00000159176	cysteine and glycine..
ENSG00000116016	endothelial PAS doma..
ENSG00000164251	F2R like trypsin rec..
ENSG00000124766	SRY-box transcriptio..
ENSG00000124762	cyclin dependent kin..
ENSG00000106366	serpin family E memb..
ENSG00000188153	collagen type IV alp..
ENSG00000122861	plasminogen activato..

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$padj),]
write.csv(res, file = "deseq_results.csv")
```

Pathway Analysis

KEGG, GO and REACTOME

```
library(pathview)
```

```
#####  
Pathview is an open source software package distributed under GNU General  
Public License version 3 (GPLv3). Details of GPLv3 is available at  
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to  
formally cite the original Pathview paper (not just mention it) in publications  
or products. For details, do citation("pathview") within R.
```

```
The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG  
license agreement (details at http://www.kegg.jp/kegg/legal.html).  
#####
```

```
library(gage)
```

```
library(gageData)  
  
data(kegg.sets.hs)  
data(sigmet.idx.hs)  
  
# Focus on signaling and metabolic pathways only  
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]  
  
# Examine the first 3 pathways  
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"  
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"  
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"  
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"  
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"  
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"  
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100"      "10201"    "10606"    "10621"    "10622"    "10623"    "107"      "10714"
[9] "108"      "10846"    "109"      "111"      "11128"    "11164"    "112"      "113"
[17] "114"      "115"      "122481"   "122622"   "124583"   "132"      "158"      "159"
[25] "1633"     "171568"   "1716"     "196883"   "203"      "204"      "205"      "221823"
[33] "2272"     "22978"    "23649"    "246721"   "25885"    "2618"     "26289"    "270"
[41] "271"      "27115"    "272"      "2766"     "2977"     "2982"     "2983"     "2984"
[49] "2986"     "2987"     "29922"    "3000"     "30833"    "30834"    "318"      "3251"
[57] "353"      "3614"     "3615"     "3704"     "377841"   "471"      "4830"     "4831"
[65] "4832"     "4833"     "4860"     "4881"     "4882"     "4907"     "50484"    "50940"
[73] "51082"    "51251"    "51292"    "5136"     "5137"     "5138"     "5139"     "5140"
[81] "5141"     "5142"     "5143"     "5144"     "5145"     "5146"     "5147"     "5148"
[89] "5149"     "5150"     "5151"     "5152"     "5153"     "5158"     "5167"     "5169"
[97] "51728"    "5198"     "5236"     "5313"     "5315"     "53343"    "54107"    "5422"
[105] "5424"     "5425"     "5426"     "5427"     "5430"     "5431"     "5432"     "5433"
[113] "5434"     "5435"     "5436"     "5437"     "5438"     "5439"     "5440"     "5441"
[121] "5471"     "548644"   "55276"    "5557"     "5558"     "55703"    "55811"    "55821"
[129] "5631"     "5634"     "56655"    "56953"    "56985"    "57804"    "58497"    "6240"
[137] "6241"     "64425"    "646625"   "654364"   "661"      "7498"     "8382"     "84172"
[145] "84265"    "84284"    "84618"    "8622"     "8654"     "87178"    "8833"     "9060"
[153] "9061"     "93034"    "953"      "9533"     "954"      "955"      "956"      "957"
[161] "9583"     "9615"
```

```
keggres = gage(res$log2FoldChange, gsets = kegg.sets.hs, ref = NULL)
head(keggres$less)
```

	p.geomean	stat.mean	p.val	q.val
hsa00232 Caffeine metabolism	NA	NaN	NA	NA
hsa00983 Drug metabolism - other enzymes	NA	NaN	NA	NA
hsa00230 Purine metabolism	NA	NaN	NA	NA
hsa04514 Cell adhesion molecules (CAMs)	NA	NaN	NA	NA
hsa04010 MAPK signaling pathway	NA	NaN	NA	NA
hsa04012 ErbB signaling pathway	NA	NaN	NA	NA

	set.size	expl
hsa00232 Caffeine metabolism	0	NA
hsa00983 Drug metabolism - other enzymes	0	NA
hsa00230 Purine metabolism	0	NA
hsa04514 Cell adhesion molecules (CAMs)	0	NA
hsa04010 MAPK signaling pathway	0	NA
hsa04012 ErbB signaling pathway	0	NA

```
keggrespathways <- rownames(keggres$greater)[1:5]
```

```
keggresids = substr(keggrespathways, start=1, stop=8)  
keggresids
```

```
[1] "hsa00232" "hsa00983" "hsa00230" "hsa04514" "hsa04010"
```

```
names(res$log2FoldChange) <- res$entrez
```

```
pathview(gene.data=res$log2FoldChange, pathway.id=keggresids, species="hsa")
```

Info: Downloading xml files for hsa00232, 1/1 pathways..

Info: Downloading png files for hsa00232, 1/1 pathways..

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa00232.pathview.png

Info: Downloading xml files for hsa00983, 1/1 pathways..

Info: Downloading png files for hsa00983, 1/1 pathways..

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa00983.pathview.png

Info: Downloading xml files for hsa00230, 1/1 pathways..

Info: Downloading png files for hsa00230, 1/1 pathways..

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa00230.pathview.png

Warning in cbind(blk.ind, j): number of rows of result is not a multiple of vector length (arg 2)

Warning in cbind(blk.ind, j): number of rows of result is not a multiple of vector length (arg 2)

Warning in cbind(blk.ind, j): number of rows of result is not a multiple of vector length (arg 2)

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Warning in cbind(blk.ind, j): number of rows of result is not a multiple of vector length (arg 2)

Warning in cbind(blk.ind, j): number of rows of result is not a multiple of vector length (arg 2)

Info: Downloading xml files for hsa04514, 1/1 pathways..

Info: Downloading png files for hsa04514, 1/1 pathways..

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa04514.pathview.png

Info: Downloading xml files for hsa04010, 1/1 pathways..

Info: Downloading png files for hsa04010, 1/1 pathways..

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa04010.pathview.png

Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
keggres_less_pathways <- rownames(keggres$less)[1:5]
keggres_less_ids = substr(keggres_less_pathways, start=1, stop=8)
names(res$log2FoldChange) <- res$entrez
pathview(gene.data=res$log2FoldChange, pathway.id=keggres_less_ids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa00232.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa00983.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa00230.pathview.png

Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
vector length (arg 2)
Warning in cbind(blk.ind, j): number of rows of result is not a multiple of
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vector length (arg 2)

'select()' returned 1:1 mapping between keys and columns

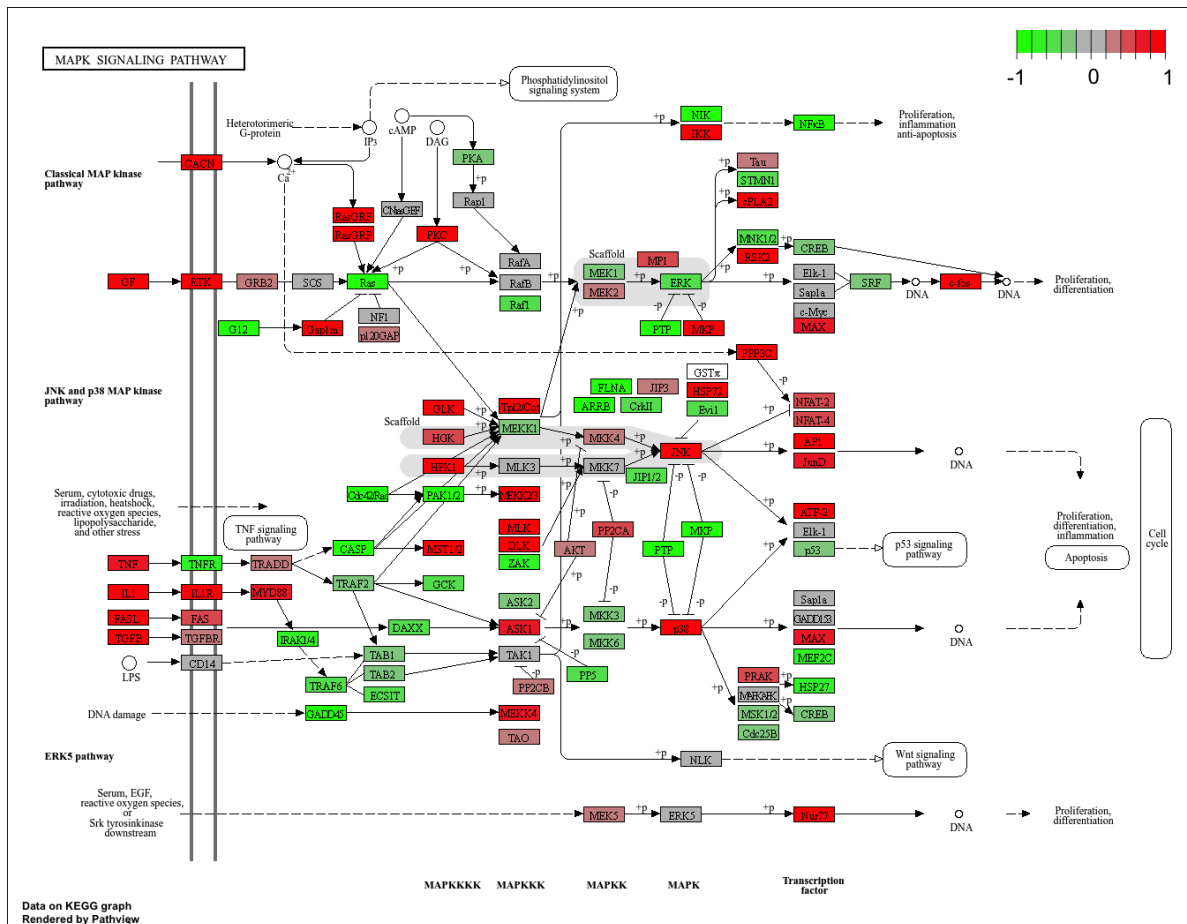
Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa04514.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/kaichristman/Desktop/Bimm 143 Spring/Class14

Info: Writing image file hsa04010.pathview.png



REACTOME

There is an R package for this analysis and a new-ish website: <https://reactome.org>

To use the website you can paste or upload a list of your DEGs

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=)
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The most significant “Entities p-value” pathway is “Cell Cycle, Mitotic” with a p-value of 2.63×10^{-5} . The most significant pathways listed in the REACTOME analysis do not match the KEGG results, which could be due to differences in the underlying databases, pathway definitions, and algorithms used for enrichment analysis. Additionally, KEGG focuses on metabolic and signaling pathways, while REACTOME includes a broader range of biological processes, which may lead to different significant pathways being identified.